

Youth Unemployment Analysis App Screenshots

<

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Deploy

Youth Unemployment Analysis in Sri Lanka

Overview

This interactive application analyses Youth Unemployment in Sri Lanka (Ages 15–29) using statistical inference and predictive modelling techniques.

Key Indicators

Total Youth Observed	Total Variables	Total Unemployed	Unemployment Rate
1428	24	189	13.2%

What This Application Provides

✓ Interactive data exploration with filters

✓ Visual analysis of unemployment patterns

✓ Statistical hypothesis testing

✓ Logistic regression prediction model

✓ Policy-relevant interpretation in simple language

<

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Deploy

✓ Visual analysis of unemployment patterns

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Data Source

Sri Lanka Labour Force Survey (LFS) – 2023

Department of Census and Statistics (DCS), Sri Lanka

Key Definitions

Youth: Individuals aged 15–29 years.

Unemployed: Persons actively seeking work and available for employment.

Labour Force: Economically active population (employed + unemployed).

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Youth Unemployment Analysis in Sri Lanka

Data Explorer

Data & Preprocessing

Data preprocessing including handling missing values, handling duplicates, variable recoding, and feature engineering (EDU_group and PROVINCE creation) was performed in Jupyter Notebook.

The cleaned dataset was exported and loaded into this Streamlit application.

Additional preprocessing such as encoding and dummy variable creation is performed dynamically within the application for modelling purposes.

Data Quality Check

Missing Values

	0
MONTH	0
SECTOR	0

Data Types

	0
MONTH	int64
SECTOR	object

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Data Quality Check

Missing Values

	0
MONTH	0
SECTOR	0
DISTRICT	0
PSU	0
HUNIT	0
HHOLD	0
SERNO	0
SEX	0
BYEAR	0
AGE	0

Data Types

	0
MONTH	int64
SECTOR	object
DISTRICT	object
PSU	int64
HUNIT	int64
HHOLD	int64
SERNO	int64
SEX	object
BYEAR	int64
AGE	int64

Filter Data

Select Sector

Urban

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Filter Data

Select Sector

Urban

Select Sex

Male

	MONTH	SECTOR	DISTRICT	PSU	HUNIT	HHOLD	SERNO	SEX	BYEAR	AGE	EDU	CJEDU	P21	Q2
0	1	Urban	Colombo	1	2	1	2	Male	1993	29	Passed G.C.E.(A/L)/ H.N.C.E.	Does not attend	Not studying	
3	1	Urban	Colombo	5	8	1	2	Male	1995	27	Passed G.C.E.(A/L)/ H.N.C.E.	Does not attend	Not studying	
8	1	Urban	Colombo	13	3	1	3	Male	2000	22	Passed Grade 10	Does not attend	Not studying	
9	1	Urban	Colombo	13	7	1	4	Male	1999	23	Passed Grade 10	Does not attend	Not studying	
11	1	Urban	Colombo	16	5	1	3	Male	1993	29	Passed G.C.E.(A/L)/ H.N.C.E.	Other Educational Institution	Currently studying/training	

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Youth Unemployment Analysis in Sri Lanka

Exploratory Analysis

Sex-wise Distribution of Unemployed Youth (%)

Female
Male

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

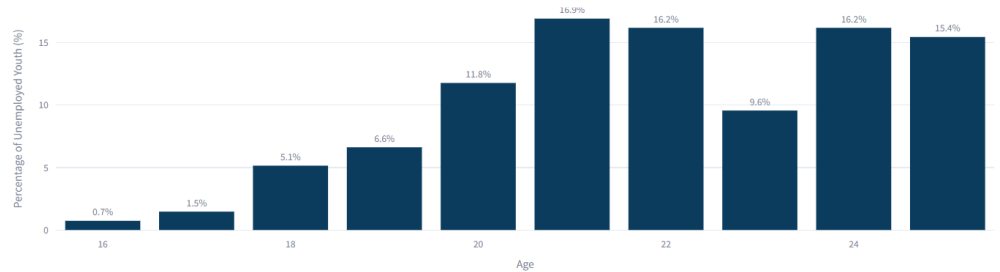
Interpretation: The pie chart shows that youth unemployment is slightly higher among females (53.4%) compared to males (46.6%). This suggests that young women may experience more difficulty finding jobs than young men, possibly due to social expectations, family responsibilities, restricted mobility, or limited access to suitable employment opportunities.

Age Distribution of Unemployed Youth (%)

Select Age Range



Age Distribution of Unemployed Youth



Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

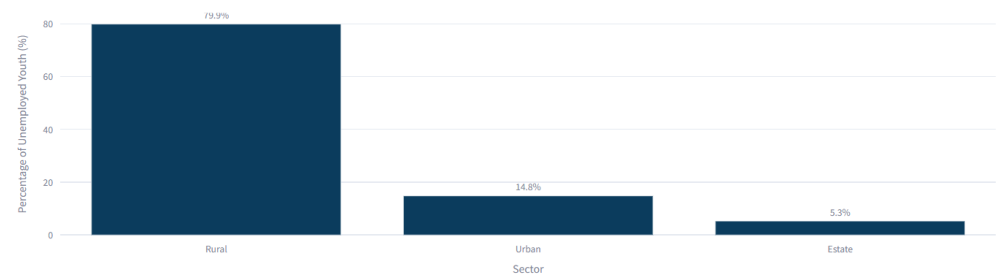
Logistic Regression

Conclusion

Interpretation: The graph shows that youth unemployment in Sri Lanka is highest among the youngest age group, especially those aged 19–24. This is mainly because many young people enter the labour market during this period after completing school or higher education and often struggle to find jobs that match their skills. As age increases toward 29, unemployment steadily declines, suggesting that with time, experience, and skill development, young people are more likely to secure employment or adjust their expectations to available opportunities.

Youth Unemployment by Sector (%)

Youth Unemployment by Sector



Interpretation: The sector-wise bar chart shows that the majority of unemployed youth are from the rural sector (79.9%), while a much smaller proportion is from urban areas (14.8%) and estate sectors (5.3%). This shows a clear gap between rural and urban areas, where young people in rural communities struggle more to find jobs because there are fewer industries and limited employment opportunities available to them.

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

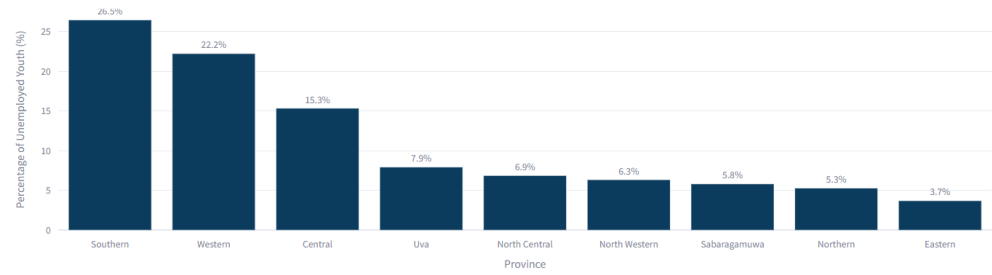
Logistic Regression

Conclusion

Youth Unemployment by Province (%)

Deploy

Youth Unemployment by Province



From the information presented in the bar chart provided above, it can be observed that there is a significant disparity in the unemployment rate of the youth in the different provinces of Sri Lanka, where there is a large percentage of unemployment in the southern and western parts of the country. The Southern province is facing the most critical unemployment situation with a 26.5% unemployment rate, while the Western province is second in the list with a 22.2% unemployment rate. On the other hand, the unemployment rate in the Eastern and Northern provinces is relatively very low, as low as 3.7%.

Filter Options

Select Sector for Gender and Age

Urban Rural Estate

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

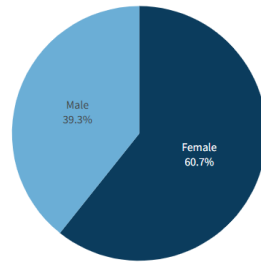
Hypothesis Testing

Logistic Regression

Conclusion

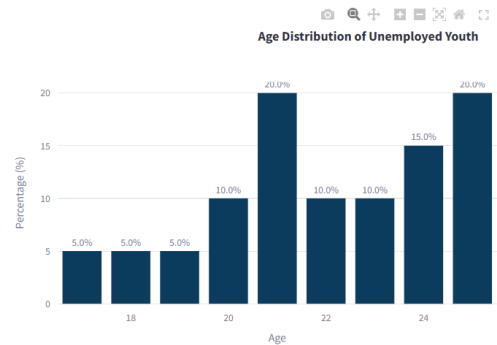
Sex-wise Distribution of Unemployed Youth (%)

Age Distribution of Unemployed Youth (%)



Female

Male



Interpretation: Within the selected sector (Urban), unemployment is higher among females, with a difference of approximately 21.4%.

Interpretation: The highest unemployment percentage occurs at age 21, accounting for approximately 20.0% within the selected range.

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Youth Unemployment Analysis in Sri Lanka

Hypothesis Testing – Statistical Inference

This section examines whether youth unemployment is statistically associated with demographic characteristics using formal hypothesis testing.

We test whether observed differences are statistically significant or occurred by random chance.

1 Chi-Square Test of Independence

This test checks whether employment status is associated with a selected categorical variable.

Select Variable to Test

SEX

Hypotheses

H_0 (Null Hypothesis): Employment status is NOT associated with SEX.

H_1 (Alternative Hypothesis): Employment status IS associated with SEX.

Navigation

- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing**
- Logistic Regression
- Conclusion

Contingency Table

SEX	Employed	Unemployed
Female	435	101
Male	804	88

Test Results

Chi-Square Statistic	p-value
22.725	0.0
Degrees of Freedom	
1	

Decision Rule

Significance Level (α) = 0.05

Since p-value < 0.05, we reject H_0 .
There is a statistically significant association between employment status and SEX.

Navigation

- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing**
- Logistic Regression
- Conclusion

2 Independent Sample t-Test (Age Difference)

This test checks whether the average age differs significantly between employed and unemployed youth.

Hypotheses

H_0 (Null Hypothesis): The mean age of employed and unemployed youth is equal.
 H_1 (Alternative Hypothesis): The mean age of employed and unemployed youth is different.

Sample Means

Mean Age (Employed)	Mean Age (Unemployed)
24.31	23.49

Test Results

t-Statistic	p-value
3.327	0.00101

Decision Rule

Significance Level (α) = 0.05

Since p-value < 0.05, we reject H_0 .
There is a statistically significant difference in mean age between employed and unemployed youth.

Overall Interpretation

These statistical tests help confirm whether the patterns observed in the exploratory analysis are statistically significant.
Significant results suggest that demographic characteristics such as sex, sector, province, and age play an important role in youth unemployment.

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Deploy

Youth Unemployment Analysis in Sri Lanka

Logistic Regression Model

Logistic Regression is used to estimate the probability that a youth is unemployed.

Model Prediction:

- 0 → Employed
- 1 → Unemployed

The model helps identify how demographic and regional factors influence youth unemployment.

Class Distribution in Dataset

	Employment Status	Count
0	Employed	1239
1	Unemployed	189

Out of 1428 youth, 189 (13.2%) are unemployed.

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Deploy

Model Performance Summary

Overall Accuracy

63.40%

Recall (Unemployed)

57.89%

Precision (Unemployed)

19.88%

Interpretation:

- Accuracy shows overall correctness of predictions.
- Recall (Unemployed) shows how well the model identifies unemployed youth.
- Precision shows how reliable unemployment predictions are.

Try Your Own Prediction

Select Age

1529

Select Sex

Male

Select Sector

Urban

Select Education Group

G.C.E.(A/L)

Youth Unemployment App

Navigation

Home

Data Explorer

Exploratory Analysis

Hypothesis Testing

Logistic Regression

Conclusion

Deploy

Model Performance Interpretation

The model achieved an accuracy of approximately 63.4%. This moderate predictive performance suggests that youth unemployment is influenced by complex structural and socio-economic factors beyond the variables included in this model (age, sex, education, sector, and province).

Therefore, while the model provides useful insights into key predictors, it does not fully capture all determinants of youth unemployment.

Model Evaluation – ROC Curve & AUC

This section adds a small new analysis to evaluate the logistic regression model. We use the ROC curve and compute the AUC to measure how well the model distinguishes between employed and unemployed youth.

AUC Score

0.664

Youth Unemployment App

Navigation

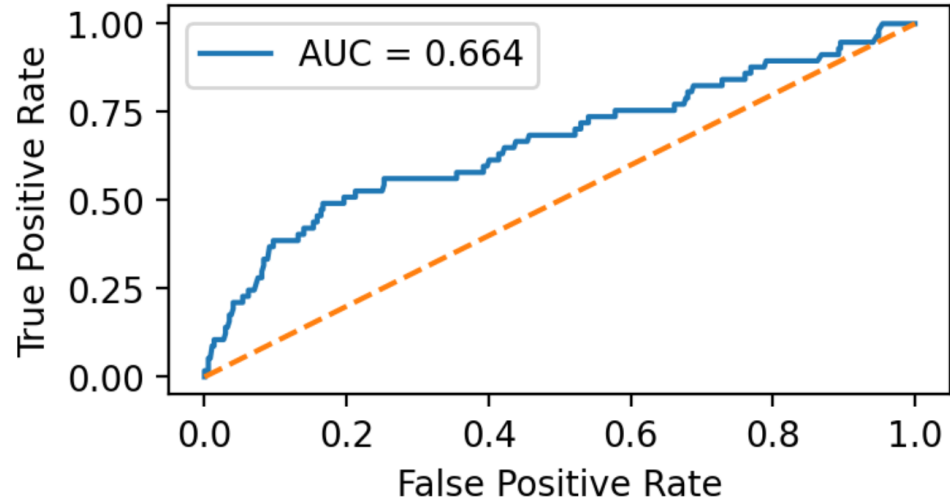
- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing

Logistic Regression

- Conclusion

Deploy

ROC Curve



False Positive Rate

Youth Unemployment App

Navigation

- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing

Logistic Regression

- Conclusion

Interpretation

The ROC curve shows how well the logistic regression model can distinguish between unemployed and employed youth.

- The AUC (Area Under the Curve) score ranges from 0 to 1:
 - 0.5 → The model is guessing randomly.
 - Closer to 1 → The model has better discrimination ability.
- Here, the AUC value indicates the model performs better than random guessing, meaning it has a moderate ability to separate unemployed from employed youth.

On the ROC graph:

- True Positive Rate (TPR)** represents the percentage of unemployed youth correctly predicted as unemployed.
- False Positive Rate (FPR)** represents the percentage of employed youth incorrectly predicted as unemployed.

In simple terms:

- A higher TPR is desirable because it means the model successfully identifies more unemployed youth.
- A lower FPR is desirable because it reduces incorrect classification of employed youth as unemployed.

Although the model is not perfect, the ROC analysis shows that it captures meaningful patterns in the data and provides additional evaluation beyond accuracy.

Deploy

Youth Unemployment App

Navigation

- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing
- Logistic Regression

Conclusion

Youth Unemployment Analysis in Sri Lanka

Conclusion & Key Insights

This web application explored youth unemployment in Sri Lanka using exploratory data analysis, hypothesis testing, and logistic regression modelling.

Key Statistical Findings

- Chi-square tests showed that employment status is significantly associated with key demographic characteristics such as sex, sector, and province.
- The independent sample t-test indicated whether there is a significant difference in the average age between employed and unemployed youth.
- These findings confirm that youth unemployment is not randomly distributed across the population.

Youth Unemployment App

Navigation

- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing
- Logistic Regression

Conclusion

- Home
- Data Explorer
- Exploratory Analysis
- Hypothesis Testing
- Logistic Regression

Conclusion

Logistic Regression Model Insights

The logistic regression model was used to estimate the probability of being unemployed based on age, sex, education level, sector, and province.

The model achieved moderate predictive accuracy. While it provides useful insights into important predictors, it does not fully capture all structural and economic factors influencing youth unemployment.

This suggests that youth unemployment is influenced by broader socio-economic conditions beyond the variables included in this dataset.

Policy Implications

- Targeted employment programs may be necessary for specific provinces or sectors.
- Education-to-employment transition policies should be strengthened.
- Gender-sensitive labour policies may help reduce disparities in youth employment.

These results provide evidence-based support for policy planning and labour market interventions.

Limitations

- The dataset includes only selected demographic variables.
- Other important factors such as income level, work experience, household background, and macroeconomic conditions were not included.

Limitations

- The dataset includes only selected demographic variables.
- Other important factors such as income level, work experience, household background, and macroeconomic conditions were not included.
- The model accuracy indicates that youth unemployment is a complex issue requiring broader structural analysis.

Overall, this application demonstrates how statistical methods and predictive modelling can be used to understand youth unemployment patterns and support data-driven decision-making in Sri Lanka.